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MLOps - Group 10

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1. Task / Problem Description

Credit risk detection is an important task in financial analysis, involving the identification of companies that may face financial distress, liquidity problems, or eventual default. Early detection of deteriorating credit quality is valuable for investors, lenders, and regulators, as warning signals often emerge well before a company reaches a critical financial state.

Traditional credit risk assessment methods rely heavily on structured financial indicators such as leverage ratios, profitability measures, and liquidity metrics. One of the most widely used approaches is the Altman Z-score, which combines accounting-based financial ratios into a single distress score. While effective as a quantitative benchmark, such methods are inherently limited to structured numerical information and may fail to capture qualitative warning signals embedded within corporate disclosures.

SEC 10-K annual filings contain rich information relevant to credit assessment, including discussion of business risks, debt obligations, operational uncertainty, and management outlook. Sections such as Item 1A (Risk Factors), Item 7 (Management Discussion and Analysis), and Item 8 (Financial Statements) often contain early indicators of financial deterioration that may not yet be fully reflected in headline financial ratios. However, manually reviewing lengthy annual reports is time-consuming, difficult to scale, and subject to analyst interpretation.

Models such as Mistral 7B can reason over long-form financial disclosures and generate structured assessments from both qualitative and quantitative evidence. However, challenges remain in grounding generated outputs to source documents and adapting general-purpose models to domain-specific financial reasoning.

The core problem addressed in this project is therefore:

> **Can retrieval augmentation and parameter-efficient fine-tuning improve LLM-based credit risk detection from SEC 10-K filings compared to direct prompting, and how do these approaches compare against the traditional Altman Z-score benchmark?**

To investigate this, we evaluate three modelling strategies:

- **Direct prompting**, where the model analyses extracted filing sections directly;
- **Retrieval-Augmented Generation (RAG)**, where relevant filing chunks are retrieved from a vector database to ground generation;
- **LoRA fine-tuning**, where lightweight adapter layers are trained to improve financial reasoning without full model retraining.

  The goal is to understand whether retrieval grounding or parameter-efficient fine-tuning provides meaningful improvements in credit risk detection, while maintaining practical scalability and interpretability.

2. Methodology

2.1 Dataset

This project uses SEC 10-K annual filings as the primary data source for credit risk assessment. A curated dataset of publicly listed U.S. companies was constructed to represent a broad spectrum of financial health conditions, ranging from financially stable firms to companies exhibiting genuine signs of distress. Each company was assigned a **binary default label** (distressed / non-distressed), a **categorical risk label** (low / medium / high), and a **continuous credit risk score (0–100)**, allowing evaluation using both threshold-based classification and continuous risk scoring.

The dataset was intentionally designed to include variation across industries and credit profiles. Low-risk companies include financially strong, investment-grade firms such as Apple, Texas Instruments, Home Depot, and UnitedHealth Group. Medium-risk firms include more cyclical or moderately leveraged businesses such as Delta Air Lines, Ford Motor Company, and Charter Communications. Distressed firms include companies with clear credit deterioration signals such as American Airlines, GameStop, AMC Entertainment, Paramount Global, and Fannie Mae.

A total of **33 companies** were included in the final dataset, comprising **10 low-risk**, **16 medium-risk**, and **7 high-risk / distressed** firms. Under binary classification, this corresponds to **7 default-positive** and **26 default-negative** companies, producing a naturally imbalanced setting representative of real-world credit distress prediction.

2.1.1 Label Construction

Ground-truth labels were manually curated using a structured financial assessment framework rather than arbitrary categorisation. Label assignment considered multiple dimensions of company credit quality:

- **Credit ratings**, using external agency ratings (e.g., S&P / Moody's) as a primary signal of creditworthiness;
- **Debt servicing ability**, focusing on leverage relative to earnings and operating cash flow rather than raw debt size alone;
- **Financial distress indicators**, including going-concern language, restructuring activity, covenant pressure, impairments, and conservatorship or bailout dependence;
- **Business model stability**, accounting for cyclicality and sector-level vulnerability (e.g., airlines, automotive, and energy sectors);
- **Default likelihood**, where binary distressed labels were assigned only when genuine default or severe deterioration risk was present.

This process produces labels that more closely reflect practical credit analysis than simple accounting-ratio thresholds, while still maintaining consistency across the dataset.

2.1.2 Section Selection

To focus analysis on financially relevant disclosures, three sections were extracted from each filing:

- **Item 1A – Risk Factors**, containing disclosed operational and financial risks;
- **Item 7 – Management's Discussion and Analysis (MD&A)**, containing management commentary on business performance, outlook, and financial drivers;
- **Item 8 – Financial Statements and Supplementary Data**, containing audited accounting disclosures and quantitative financial metrics.

These sections combine qualitative management disclosure with quantitative financial evidence, making them particularly suitable for credit risk modelling.

2.2 Data Preprocessing

To prepare filings for modelling, a multi-stage preprocessing pipeline was developed to convert raw annual reports into structured, searchable financial text suitable for prompting, retrieval, and fine-tuning tasks.

2.2.1 PDF Text Extraction

Each annual report was first converted from PDF format into raw machine-readable text using PyMuPDF. This extraction stage preserves the full filing content while removing dependence on manual section copy-pasting. Since SEC filings vary in formatting and layout, extracted text often contains structural noise such as repeated headers, page numbering artifacts, and formatting inconsistencies, requiring further cleaning downstream.

2.2.2 Section Extraction

The preprocessing pipeline isolates the three target sections — Item 1A, Item 7, and Item 8 — using a rule-based section parser designed to identify section boundaries within extracted filing text.

Care was taken to avoid false matches caused by table-of-contents references, repeated section headers, and formatting inconsistencies appearing throughout the document. Final extracted sections were stored as structured per-company JSON files, forming the primary textual input for all downstream modelling approaches.

2.2.3 Text Cleaning and Chunking

Extracted section text was normalised and segmented into overlapping chunks to support retrieval and embedding generation. Chunking was performed using a fixed window size of **512 tokens** with **64-token overlap** between neighbouring chunks.

This chunking strategy serves two purposes:

1. **Handling long financial disclosures**, which exceed model context limits when processed in full;
2. **Preserving local context**, where overlap ensures important sentences near chunk boundaries are not fragmented or lost.

Each chunk retains metadata including:

- company ticker,
- company name,
- filing section, and
- chunk identifier.

This metadata is later used for retrieval filtering and citation grounding.

2.2.4 Embedding Generation and Vector Indexing

To support semantic retrieval, each text chunk was converted into a dense vector embedding using the transformer embedding model **BAAI/bge-base-en-v1.5**. Unlike keyword matching, dense embeddings capture semantic similarity between financial concepts such as liquidity pressure, debt burden, restructuring risk, and weakening operating performance.

Generated embeddings were stored in a FAISS vector index, enabling efficient nearest-neighbour search during inference. Retrieved chunks were further filtered using company ticker and filing section metadata to improve company-specific retrieval precision and reduce semantically similar but irrelevant matches across companies.

In total, the preprocessing pipeline transforms raw SEC filings into structured financial sections, chun

2.3 Data Splitting Strategy

Given the relatively small labelled dataset, this project uses **stratified 5-fold cross-validation** to provide robust model evaluation while maintaining consistent class balance across splits. Stratification preserves the distribution of low-risk, medium-risk, and distressed companies in each fold, ensuring that every partition remains representative of the overall dataset.

Under this setup, the 33-company dataset is divided into five approximately equal folds. For each iteration, one fold is held out as the **test split**, while the remaining folds are used for **training and validation**. This process is repeated across all five folds so that each company appears in the held-out test set exactly once, while never appearing in both training and testing within the same fold.

Within each fold:

- the **training split** is used for LoRA adapter optimisation,
- the **validation split** is used for hyperparameter selection and training monitoring, and
- the **held-out test split** is used for evaluation.

For the **baseline**, **RAG**, and **LoRA** approaches, inference is performed only on companies in the held-out test split for that fold. This ensures that every company is evaluated under the same standardized protocol and that no model is assessed on companies seen during its corresponding training process.

Using stratified cross-validation provides several advantages:

- **More reliable performance estimates** than a single fixed train-test split on a small dataset;
- **Consistent class balance** across folds;
- **Reduced overfitting and leakage risk**, since companies are never evaluated on models trained using their own labels within the same fold;
- **Fair comparison across approaches**, as baseline prompting, retrieval-augmented generation, and LoRA fine-tuning are all evaluated on identical data partitions.

Final reported metrics are aggregated across all folds to provide an overall estimate of model performance.

2.4 Prompt Design

To ensure consistency across modelling approaches and enable automated evaluation, prompt templates were designed to produce structured credit risk assessments using a common output schema. This schema was used for both **baseline prompting**, **retrieval-augmented generation (RAG)**, and as the target output format during **LoRA fine-tuning**, ensuring comparable outputs across all model variants.

Each model output contains the following components:

- **Continuous risk score (0–100):** a standardized numerical estimate of company credit risk;
- **Categorical risk label (low / medium / high):** an interpretable qualitative classification of overall risk level;
- **Key signals:** the primary financial warning signs or supporting indicators identified in the filing;
- **Citations / retrieved references:** links to relevant source passages or retrieved chunks used to support the prediction;
- **Reasoning:** a concise explanation summarizing the basis of the model's credit risk judgement.

Prompt instructions explicitly guide the model to assess company creditworthiness using financial disclosures extracted from SEC 10-K filings, with emphasis on operational risk exposure, debt burden, liquidity position, restructuring signals, and management commentary. For the **baseline** and **LoRA** approaches, prompts are constructed primarily from extracted **Item 1A (Risk Factors)** and **Item 7 (Management Discussion and Analysis)** sections. For the **RAG** approach, semantically retrieved passages from the indexed filing corpus are prepended to the prompt to provide additional grounding context before generation.

To support reliable downstream evaluation, all model outputs are returned in a strict **JSON** format containing predefined fields (risk_score, risk_level, key_signals, citations, and reasoning). Standardising outputs in this way reduces response variability, enables automated parsing of predictions, and supports consistent benchmarking across modelling approaches.

An example prompt instruction is shown below:

> *Assess the credit risk of the company based on the provided financial disclosures. Return your answer as a JSON object containing the fields: risk_score, risk_level, key_signals, citations, and reasoning.*

By enforcing a common structured output format, the system improves interpretability, reproducibility, and consistency in model evaluation across baseline, retrieval-augmented, and fine-tuned approaches.

3. Experiments

3.1 Models and Baselines

To evaluate the effectiveness of different modelling strategies for credit risk detection, this project compares four approaches: a traditional financial distress benchmark, a direct prompting baseline, a retrieval-augmented generation pipeline, and a parameter-efficient fine-tuned model. All approaches are evaluated using the standardized stratified cross-validation framework described in Section 2.3, ensuring consistent comparison across identical data partitions.

3.1.1 Altman Z-score Baseline

As a traditional benchmark, this project uses the Altman Z-score, a widely used financial distress prediction model based on accounting ratios. The model combines five financial indicators into a single distress score:

$Z = 1.2X_1 + 1.4X_2 + 3.3X_3 + 0.6X_4 + 1.0X_5$

where:

X_1 = Working Capital / Total Assets

X_2: Retained Earnings / Total Assets

X_3: EBIT / Total Assets

X_4: Market Value of Equity / Total Liabilities

X_5: Revenue / Total Assets

The resulting score is interpreted using standard financial distress thresholds:

- **$Z < 1.81$** → distressed / high risk
- **$1.81 \leq Z \leq 2.99$** → grey zone / medium risk
- **$Z > 2.99$** → financially safe / low risk

For consistency with the LLM-based approaches, Altman Z-scores were mapped into the same low / medium / high risk categories and converted into binary distress predictions for evaluation.

The Altman Z-score provides an interpretable quantitative benchmark based entirely on structured financial statement variables, allowing comparison against text-based reasoning approaches that incorporate qualitative disclosures.

3.1.2 Baseline Prompting

The baseline prompting approach uses Mistral 7B as a direct credit risk reasoning model without retrieval augmentation or additional fine-tuning. Extracted filing sections are provided directly to the model using the structured prompt format described in Section 2.4, and the model generates a complete credit risk assessment from disclosure text alone.

Inputs are constructed primarily from:

- **Item 1A – Risk Factors**, and
- **Item 7 – Management Discussion and Analysis (MD&A)**

The model outputs:

- a continuous risk score (0–100),
- a categorical risk label (low / medium / high),
- key financial warning signals,
- supporting citations / references, and
- concise reasoning.

This approach serves as the primary LLM baseline, measuring how effectively a general-purpose language model can infer financial distress signals using prompt engineering alone.

3.1.3 Retrieval-Augmented Generation (RAG)

The Retrieval-Augmented Generation (RAG) approach extends the baseline prompting framework by incorporating semantic retrieval prior to generation. For each company, the system retrieves the most relevant financial text chunks from the indexed filing corpus using dense vector similarity search over the FAISS index.

Retrieved chunks are filtered using company ticker and section metadata before being prepended to the prompt, providing the model with targeted financial context relevant to the company being analysed. This grounding mechanism is intended to:

- reduce unsupported claims,
- improve citation quality, and
- strengthen evidence-based financial reasoning.

Compared with direct prompting, RAG introduces an explicit retrieval stage that enables the model to reason over semantically relevant disclosures rather than relying solely on truncated filing context.

3.1.4 LoRA Fine-Tuned Model

To improve domain-specific financial reasoning, this project evaluates a fine-tuned version of Mistral 7B using **Low-Rank Adaptation (LoRA)**, a parameter-efficient fine-tuning method that updates small trainable adapter matrices while keeping the original model weights frozen.

Fine-tuning was performed using **QLoRA**, combining low-rank adaptation with **4-bit NF4 quantization** to reduce memory usage while maintaining training effectiveness. This allows efficient specialization of the base model for credit risk analysis without requiring full-model retraining.

Three LoRA configurations were evaluated:

- **Rank 8 (r = 8)** — lower-capacity adaptation with fewer trainable parameters;
- **Rank 16 (r = 16)** — balanced adaptation capacity;
- **Rank 32 (r = 32)** — higher-capacity adaptation with greater representational flexibility.

This comparison allows evaluation of how adapter capacity influences domain adaptation and financial reasoning performance under limited labelled data.

3.2 Experimental Setup

All approaches were evaluated using the same cross-validation protocol and common modelling pipeline. Key experimental settings are summarized in Table 1.

Parameter	Value
Base model	Mistral 7B
Embedding model	BAAI/bge-base-en-v1.5
Vector store	FAISS
Chunk size	512 tokens
Chunk overlap	64 tokens
Cross-validation	Stratified 5-fold
Binary distress threshold	Risk score ≥ 70
LoRA configurations	r = 8, 16, 32
Quantization	4-bit NF4 (QLoRA)

Experiments were conducted in a **Google Colab environment using an NVIDIA T4 GPU**. The primary language model used in this work was **Mistral 7B Instruct v0.2**, with parameter-efficient fine-tuning performed using **LoRA adapters**. Dense retrieval for the RAG pipeline used embeddings generated from **BAAI bge-base-en-v1.5**, indexed using **FAISS**.

Across model training, fine-tuning, repeated evaluations, and iterative experimentation, the project used **approximately 30 GPU-hours in total**. Once finalized, the full experimental pipeline — including inference across all approaches and evaluation — can be executed in approximately **3–4 GPU-hours**, demonstrating relatively modest compute requirements.

3.3 Evaluation Metrics

Model performance was evaluated using both classification and grounding-based metrics.

For binary classification, continuous risk scores were converted into distress predictions using the following threshold:

$$\hat{y} = \begin{cases} 1, & \text{score} \geq 70 \\ 0, & \text{score} < 70 \end{cases}$$

where predictions with scores of **70 or above** are classified as distressed (positive class), while lower scores are classified as non-distressed.

The following metrics were used:

- **Accuracy**, measuring overall prediction correctness;
- **Precision**, measuring how often predicted distressed firms were truly distressed;
- **Recall**, measuring the proportion of distressed firms correctly identified;
- **F1-score**, balancing precision and recall;
- **ROC-AUC**, measuring ranking quality across classification thresholds.

Final reported results are aggregated across all cross-validation folds to provide an overall estimate of model performance.

4. Results and Discussion

4.1 Overall Performance

Final model performance aggregated across all stratified cross-validation folds is summarized in Table 2.

Approach	ROC-AUC	F1	Precision	Recall	Accuracy
Baseline Prompting	**0.849**	0.000	0.000	0.000	**0.813**
Retrieval-Augmented Generation (RAG)	0.692	**0.429**	0.273	**1.000**	0.500
LoRA Fine-Tuning (r = 32)	0.701	0.000	0.000	0.000	0.733
RAG + LoRA	0.500	0.000	0.000	0.000	**0.813**
Altman Z-score	0.778	**0.444**	**0.333**	0.667	0.444

Table 2. Aggregated performance across all folds and companies.

Several important findings emerge from the evaluation.

First, the **Altman Z-score achieved the strongest balanced classification performance overall**, producing the highest F1-score (0.444) and strongest precision (0.333) among all approaches while maintaining recall of 0.667. This demonstrates that traditional ratio-based financial distress modelling remains a highly competitive benchmark for credit risk detection, particularly when labelled datasets are small.

Second, among the large language model-based approaches, **Retrieval-Augmented Generation (RAG) delivered the strongest classification performance**, achieving F1-score of 0.429 and perfect recall (1.000), successfully identifying all distressed firms in the evaluation set. This indicates that retrieval grounding materially improved the model's ability to detect financial distress signals embedded in corporate disclosures.

Third, although **Baseline Prompting produced the highest ROC-AUC (0.849)**, it failed to classify any company as distressed under the evaluation threshold, producing zero precision,

zero recall, and zero F1-score. This indicates that while the baseline model ranked distressed firms higher on average than healthy firms, its predicted scores remained too conservative to cross the distress classification boundary.

Finally, **LoRA fine-tuning did not improve classification performance**, and combining retrieval augmentation with fine-tuning further degraded results, suggesting that parameter-efficient adaptation remained ineffective in this limited-data setting.

Find the logs for our runs here:
https://wandb.ai/pankti0-singapore-university-of-technology-and-design/credit-risk-detection/runs/0qak6z4z/logs?nw=nwuserpankti0

4.2 Baseline Prompting Performance

The direct prompting baseline produced the highest ROC-AUC (0.849), indicating strong ranking ability. In other words, distressed firms generally received higher predicted risk scores than financially healthy firms, suggesting that the model captured meaningful differences in financial risk profiles.

However, the model consistently assigned relatively conservative risk scores that remained below the distress threshold. As a result, it failed to generate positive distressed classifications, producing zero recall and zero F1-score.

This reveals an important distinction between **ranking quality** and **classification quality**. While the baseline model could separate distressed firms from healthy firms on a relative scoring basis, it was poorly calibrated for binary distress classification.

Several factors likely contributed to this conservative scoring behaviour:

- instruction-tuned models often avoid extreme outputs without explicit calibration;
- direct prompting lacked retrieval evidence that might strengthen distress signals;
- prompt design may have encouraged moderate rather than high-risk scoring.

Overall, baseline prompting demonstrated promising financial reasoning ability but poor classification calibration.

4.3 Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation produced the strongest overall performance among LLM-based approaches.

Most notably, RAG achieved:

- **perfect recall (1.000)**, correctly identifying all distressed firms;
- **F1-score of 0.429**, competitive with Altman Z-score;
- improved positive classification behaviour compared with direct prompting.

This suggests retrieval grounding successfully strengthened the model's sensitivity to financial distress signals. By retrieving semantically relevant filing passages prior to generation, the model was exposed to stronger evidence related to:

- liquidity pressure,
- debt burden,
- refinancing risk,
- restructuring signals,
- deteriorating operating performance.

Qualitative inspection of generated outputs also showed stronger grounding behaviour, with retrieved evidence often surfacing meaningful financial warning signals that aligned with analyst-style credit reasoning.

The main limitation of RAG was reduced precision (0.273), meaning financially healthy firms were often incorrectly flagged as distressed. This suggests the retrieval pipeline improved sensitivity but increased false positives by overweighting retrieved risk evidence.

Despite this limitation, RAG demonstrated that **retrieval grounding can materially improve distress detection performance in financial disclosure analysis**, particularly when recall is prioritized.

4.4 LoRA Fine-Tuning

LoRA fine-tuning did not improve model performance in this project.

The best-performing LoRA configuration (rank = 32) achieved ROC-AUC of 0.701 but produced zero positive distress classifications, resulting in zero precision, zero recall, and zero F1-score under the evaluation threshold.

This indicates that although the fine-tuned model retained some ranking ability, it failed to calibrate outputs toward actionable distress predictions.

Several factors likely contributed:

- extremely limited labelled training data (33 companies, only 7 distressed firms);
- severe class imbalance;

- limited diversity of distress examples;
- overfitting to fold-specific patterns;
- insufficient supervised signal for robust adapter learning.

These results suggest that parameter-efficient fine-tuning is not automatically beneficial in small-domain datasets, particularly when label scarcity limits meaningful specialization.

4.5 Combined RAG + LoRA

Combining retrieval augmentation with LoRA fine-tuning produced the weakest result overall.

The combined model achieved:

- ROC-AUC of 0.500 (equivalent to random ranking),
- zero precision,
- zero recall,
- zero F1-score.

This indicates that combining retrieval grounding with fine-tuning compounded model instability rather than improving reasoning.

Instead of complementary gains, retrieval-stage noise and limited-data fine-tuning appear to have reinforced poor score calibration, causing the model to lose both ranking ability and classification performance.

This result highlights an important practical lesson:

> **Increasing modelling complexity does not necessarily improve performance when training data is limited and retrieval quality remains imperfect.**

4.6 Key Findings

Three major findings emerge from this study:

1. **Altman Z-score remained the strongest balanced classifier**, demonstrating that structured financial ratios remain highly effective for credit distress prediction.
2. **RAG was the strongest LLM-based approach**, achieving perfect recall and competitive F1-score, showing that retrieval grounding materially improves distress detection.

3. **LoRA fine-tuning did not improve performance**, and combining LoRA with retrieval further degraded results, highlighting the limitations of parameter-efficient adaptation under limited supervision.

Overall, in this small-data credit-risk setting, **retrieval augmentation provided the most meaningful improvement for LLM-based credit distress detection**, while classical financial ratio models remained highly competitive benchmarks for practical classification performance.

4.7 Limitations

Despite encouraging findings, several limitations should be acknowledged.

First, the dataset size remains relatively small, consisting of only 33 companies with 7 distressed firms. While stratified 5-fold cross-validation improves robustness, the limited number of positive distress examples constrains statistical confidence and makes model evaluation sensitive to individual company cases.

Second, ground-truth labels were manually curated using a structured credit assessment framework rather than historical default events alone. Although this produces more realistic analyst-style credit labels, some degree of subjective judgement remains in label construction.

Third, retrieval quality remains an important bottleneck for RAG performance. While semantic retrieval successfully surfaced financially relevant disclosures, retrieved passages occasionally included noisy financial tables or weakly predictive text, which likely contributed to false positives.

Finally, the project evaluates a relatively small set of modelling configurations. More extensive prompt engineering, retrieval re-ranking, score calibration, and larger-scale domain adaptation may further improve performance beyond the configurations explored here.

4.8 Future Work

Several promising directions exist for future improvement.

First, expanding the labelled dataset with a larger and more diverse set of distressed companies would improve statistical reliability and provide stronger supervision for fine-tuning approaches such as LoRA.

Second, hybrid structured-text modelling presents a strong opportunity. Combining qualitative disclosure reasoning with structured financial variables such as leverage ratios, liquidity metrics, and Altman-style accounting indicators may improve both ranking and classification performance.

Third, retrieval quality can be improved through stronger chunk ranking, financial-domain retrieval tuning, and passage compression techniques that reduce prompt noise while preserving relevant evidence.

Finally, score calibration methods could substantially improve classification quality. As observed in the baseline prompting model, strong ranking ability does not necessarily translate into effective binary classification. Better calibration of continuous risk scores into distress predictions remains an important practical direction.

5. Conclusion

This project evaluated multiple approaches for credit risk detection from SEC 10-K annual filings, comparing traditional financial ratio analysis with large language model-based reasoning methods. Specifically, direct prompting, retrieval-augmented generation (RAG), LoRA fine-tuning, and combined RAG + LoRA were benchmarked against the classical Altman Z-score using stratified 5-fold cross-validation on a curated 33-company dataset.

Among the large language model-based approaches, **Retrieval-Augmented Generation (RAG) delivered the strongest classification performance**, achieving perfect recall (1.000) and an F1-score of 0.429, successfully identifying all distressed firms in the evaluation set. This demonstrates that retrieval grounding materially improved the model's ability to surface and act on financially relevant warning signals embedded within long-form corporate disclosures.

At the same time, the **Altman Z-score remained the strongest balanced classifier overall**, achieving the highest F1-score (0.444) and strongest precision (0.333) while maintaining competitive recall. This highlights the continued effectiveness of structured financial ratio analysis as a practical benchmark for distress prediction.

Direct prompting with Mistral 7B achieved the highest ROC-AUC (0.849), indicating strong ranking ability, but failed to generate positive distressed classifications under the chosen decision threshold. This suggests that while the base model captured meaningful relative differences in credit quality, its outputs remained conservatively calibrated for binary distress classification.

By contrast, **LoRA fine-tuning did not improve performance**, and combining LoRA with retrieval augmentation further degraded results. These findings suggest that parameter-efficient adaptation offers limited benefit when labelled financial datasets are small and class imbalance is severe.

Overall, the findings suggest that **retrieval augmentation provides the most meaningful improvement for LLM-based credit distress detection**, particularly when sensitivity to distress signals is prioritized. However, traditional structured financial models remain highly competitive, and effective credit risk modelling may ultimately benefit most from hybrid approaches that combine qualitative disclosure reasoning with structured financial indicators.